

Female sex hormones and contraception

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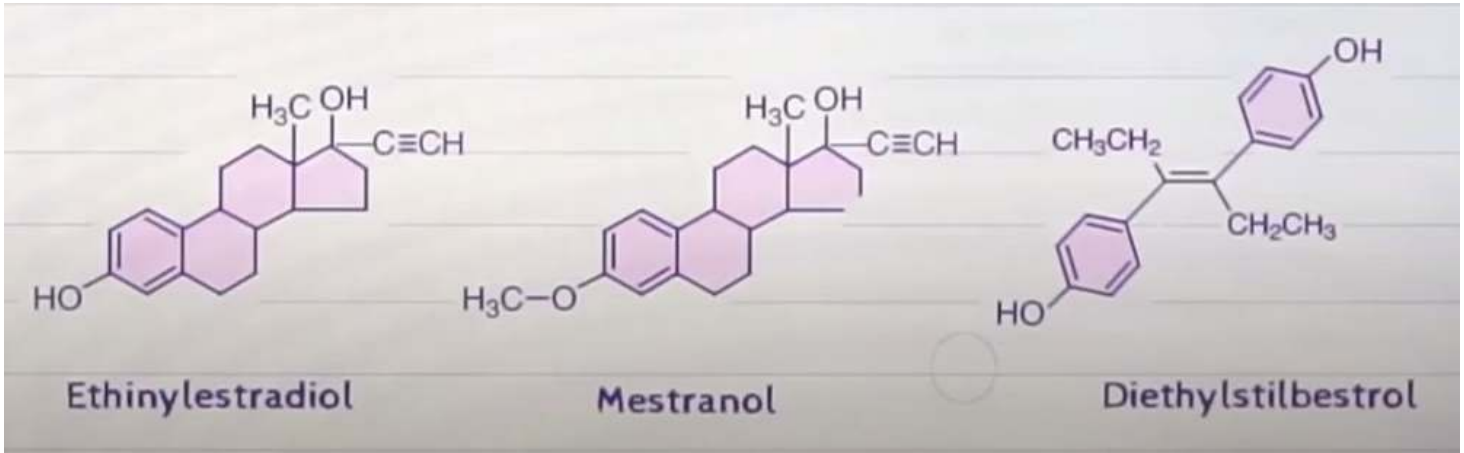
Outline

- Physiology of menstrual cycle
- Estrogen
- Progesterone
- Hormonal contraception

OESTROGENS

- Oestrogens are synthesised by the ovary, placenta, and in small amounts by the testis and adrenal cortex.
- **Preparations** Many preparations (oral, transdermal, intramuscular, implantable and topical) of oestrogens are available for a wide range of indications. These preparations include:

- **natural** :e.g. ***oestradiol, oestrone*** *and* ***estriol***
- **synthetic:** e.g. ***ethinylestradiol, mestranol, Diethylstilbestrol***
- **Conjugated estrogens:** are mixture of several estrogen hormones



Actions:

- Oestrogen induces synthesis of progesterone receptors in uterus, vagina, anterior pituitary and hypothalamus.
- Conversely, progesterone decreases estrogen receptor expression in the reproductive tract.

Physiologic effects

- Development of primary sex characters (genital tract).
- Development of ♀ secondary sex characters (fat distribution, pubic hair and breast).
- Metabolic effects:

✓ Increase bone mass and prevent bone resorption.

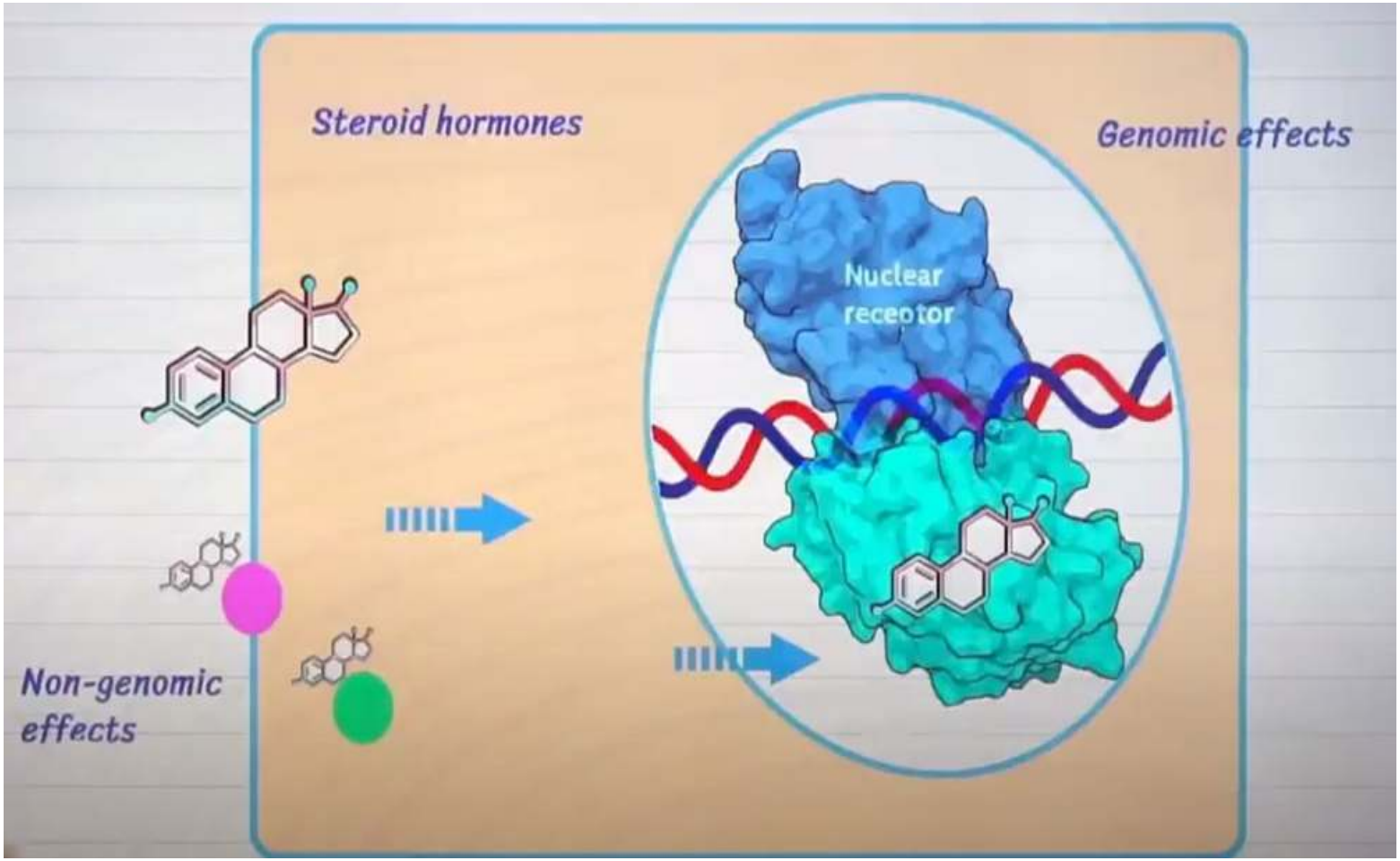
✓ Increase concentrations of *high-density lipoproteins* contribute to the relatively low risk of atheromatous disease in premenopausal women compared with men of the same age.

✓ Salt and water retention.

- Increase blood coagulation and ***increase the risk of thromboembolism***.
- Oestrogen-mediated dilation of coronary arteries occurs by the increased formation and release of *nitric oxide* and *prostacyclin* in endothelial cells.
- In CNS affect libido and mood.

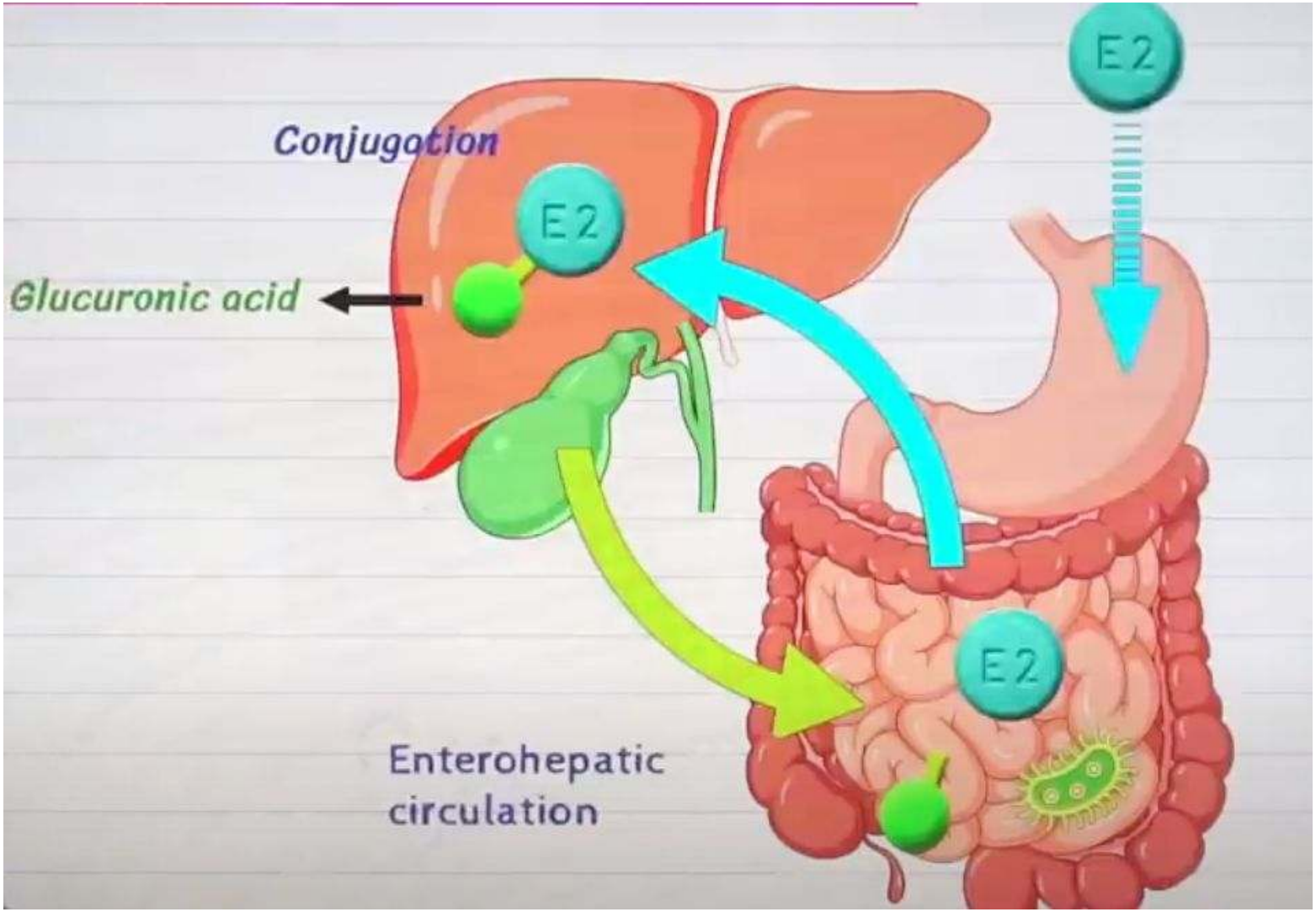
Mechanism of action

- steroid hormones bind with high affinity to ***specific nuclear oestrogen-receptor proteins*** that mediate genomic regulation
- Some physiologic oestrogen effects, are initiated by *interaction with membrane receptors*, including a G protein-coupled oestrogen receptor


Steroid hormones
Genomic effects
Nuclear receptor
Non-genomic effects

Pharmacokinetics

- Estrogens are well absorbed in the gastrointestinal tract, Readily absorbed from skin and mucous membranes.
- They are bound to **albumin** and to a sex steroid-binding **globulin**.
- They are metabolized by hydroxylation and glucuronides conjugation in the liver.
- Excreted in urine and bile, undergo enterohepatic cycling.


Conjugation
E2
Glucuronic acid
E2
E2
Enterohepatic
circulation

Clinical uses:

1. Contraceptive pills in combination with progesterone.
2. Dysfunctional uterine bleeding.
3. Replacement therapy in primary ovarian failure and delay puberty.
4. Premature menopause.
5. Postmenopausal symptoms e.g. atrophic vaginitis, hot flush and osteoporosis.

HORMONE REPLACEMENT THERAPY

- Loss of ovarian hormones after menopause can result in **'POSTMENOPAUSAL SYNDROME'** in ~50% of women.
- Symptoms: vasomotor symptoms (hot flushes or flashes), sleep disturbances, CVS problems, vaginal dryness, osteoporosis, and depression. HRT can significantly improve these sym
  - Women with removed uterus take E2 alone.
  - Women with intact uterus require E2 combined with Pg to decrease the risk of endometrial hyperplasia and cancer.

- **Side effects:**

- tenderness in the breasts,
- nausea, vomiting, anorexia,
- retention of salt and water with resultant oedema,
- increased risk of thromboembolism
- postmenopausal replacement therapy, oestrogens causes ***endometrial hyperplasia*** unless given cyclically with a progestogen.
- When administered to males, oestrogens result in feminisation.
- Oestrogen administration to pregnant women can cause genital abnormalities in their offspring.

Oestrogen antagonists:

- **Selective Oestrogen Receptor modulators (**SERMs)

  SERMs are a class of oestrogen-related compounds that display ***competitive partial agonists*** for **estrogen receptors** depending on the tissue type. includes

  tamoxifen, ***raloxifene***.

- **Oestrogen Receptor Blockers:**

  Clomiphene

- **Aromatase Inhibitors:**

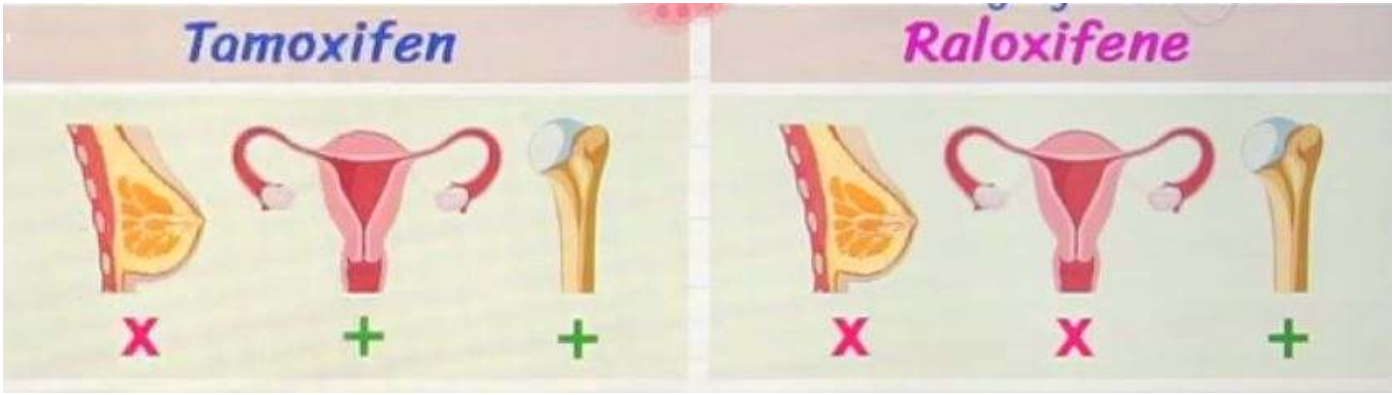
  Letrozole & Anastrazole



Raloxifene & Tamoxifen

Selective estrogen receptor modulators (SERMs)

- **Raloxifene has** anti-estrogenic effects on breast and uterus but oestrogenic effects on bone, lipid metabolism and blood coagulation.
- **Tamoxifen** has an anti-oestrogenic action on mammary tissue but oestrogenic actions on endometrium, bone plasma lipids and clotting factors.



Raloxifene

- Clinical uses: It is used for the treatment and prevention of postmenopausal osteoporosis, prophylactic therapy to reduce the risk of breast cancer in high-risk patients and dysfunctional uterine bleeding.

- Adverse effects: hot flashes and thrombotic complications.

Tamoxifen

- Clinical uses: It is used in the treatment of advanced, estrogen receptor positive breast cancer pre and post menopause and prophylactic therapy to reduce the risk of breast cancer in high-risk patients. as adjuvant therapy following mastectomy or radiation for breast cancer.

- Adverse effects: tamoxifen increases the risk of endometrial cancer and thrombotic complications.

PROGESTOGENS

- The natural progesterone is called **(progesterone**) while synthetic one is called **progestin**
- It is secreted by the corpus luteum in the second part of the menstrual cycle, and by the placenta during pregnancy. Small amounts are also secreted by the testis and adrenal cortex.
- Progestogens act, as do other steroid hormones, on nuclear receptors.
- The density of progesterone receptors is controlled by estrogens.

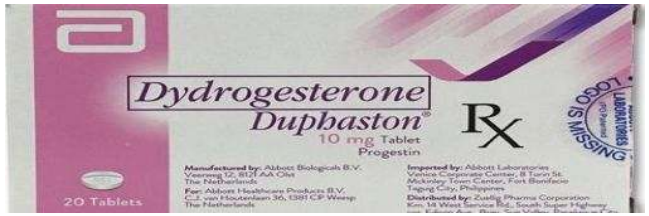
Preparations

There are three main groups of progestogens:

1. Natural progesterone available as suppository.
2. progesterone derivatives

(e.g. **hydroxyprogesterone**, **medroxyprogesterone**, **cyproterone, dydrogesterone**). Derivatives are available for oral, intramuscular injection and suppository.









3. Testosterone derivatives

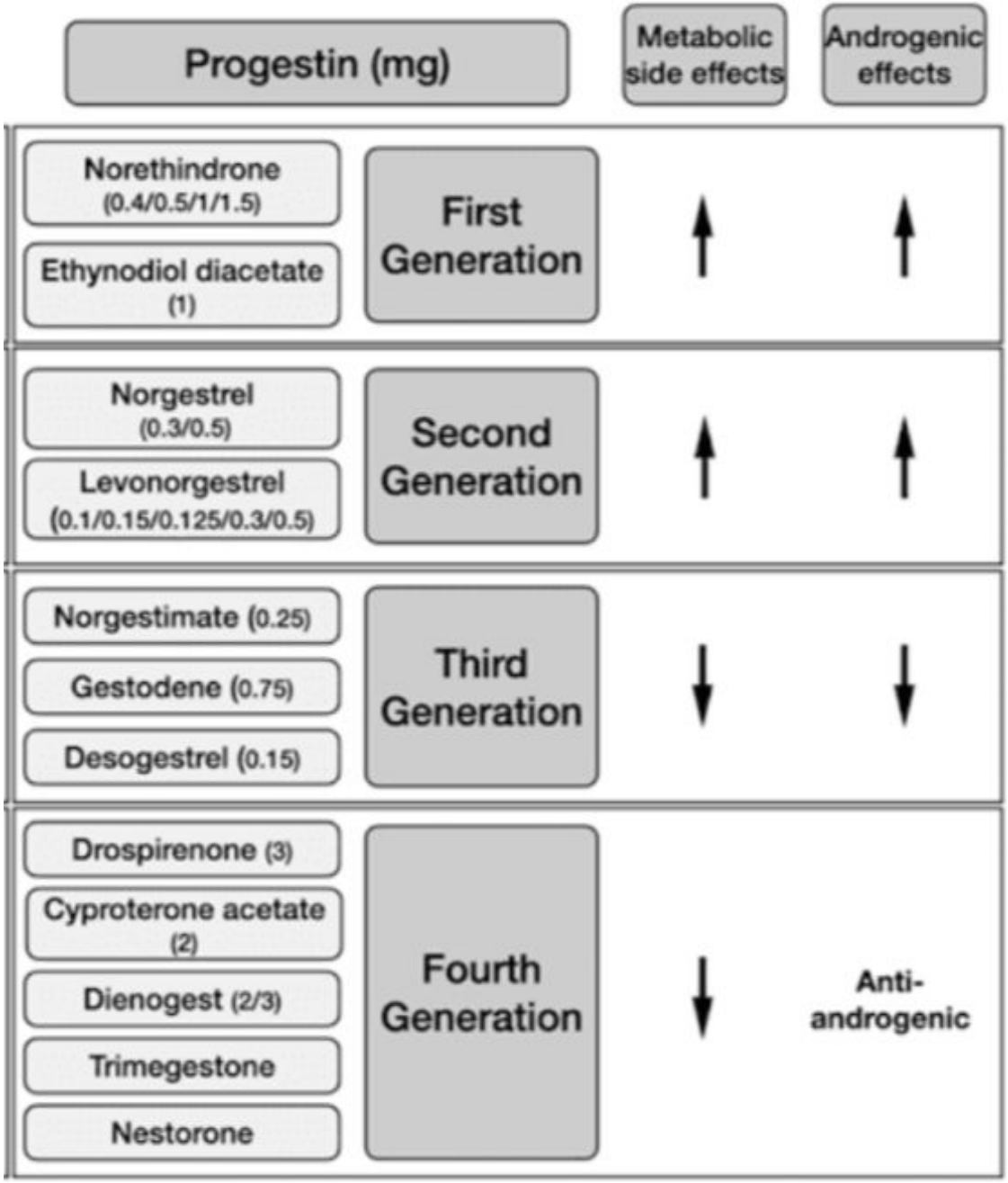
- Older progestogens (e.g. **norethisterone and norgestrel**). They have androgenic activity and are metabolized to give estrogenic products



- Newer progestogens include **desogestrel and gestodene**; they are considered for women who experience side effects such as acne, depression or breakthrough bleeding with the older drugs.

4. Aldosterone derivatives, **drospirenone**, is an analogue of the aldosterone exhibits both progestational and antiandrogenic activity indicated for women experience acne, hirsutism, nausea, and lipid changes.




Progestin (mg)
Metabolic side effects
Androgenic effects
Norethindrone (0.4/0.5/1/1.5)
Ethynodiol diacetate (1)
First Generation
Norgestrel (0.3/0.5)
Levonorgestrel (0.1/0.15/0.125/0.3/0.5)
Second Generation
Norgestimate (0.25)
Gestodene (0.75)
Desogestrel (0.15)
Third Generation
Drospirenone (3)
Cyproterone acetate (2)
Dienogest (2/3)
Trimegestone
Nestorone
Fourth Generation
Anti-androgenic

Pharmacokinetic aspects

- Injected progesterone is bound to albumin.
- Some is stored in adipose tissue.
- It is metabolized in the liver, primarily by cytochrome P450 enzymes in the liver and excreted in the urine.

Unwanted effects

acne, fluid retention, weight change, depression, change in libido, breast discomfort, premenstrual symptoms, irregular menstrual cycles and breakthrough bleeding. There is an increased incidence of thromboembolism.

Clinical uses of progestogens

Progestogens

• Contraception:

– with oestrogen in combined oral contraceptive pill

– as progesterone-only contraceptive pill

– as injectable or implantable progesterone-only contraception

– as part of an intrauterine contraceptive system.

• Combined with oestrogen for oestrogen replacement therapy in women with an intact uterus, to prevent endometrial hyperplasia and carcinoma.

- For endometriosis.
- In endometrial carcinoma.
- menstrual disorders.
- Threaten abortion and premature delivery (**natural progesterone** and **dydrogesterone**)

ANTIPROGESTOGENS

- ✓ ***Mifepristone*** is a partial agonist at progesterone receptors. It sensitizes the uterus to the action of prostaglandins.
- ✓ It is given orally .
- ✓ Mifepristone is used, in combination with a prostaglandin (***misoprostol***), as a medical alternative to surgical termination of pregnancy.
- ✓ **The major adverse effects** are

1. Abdominal pain,
2. Uterine bleeding, and
3. The possibility of an incomplete abortion.

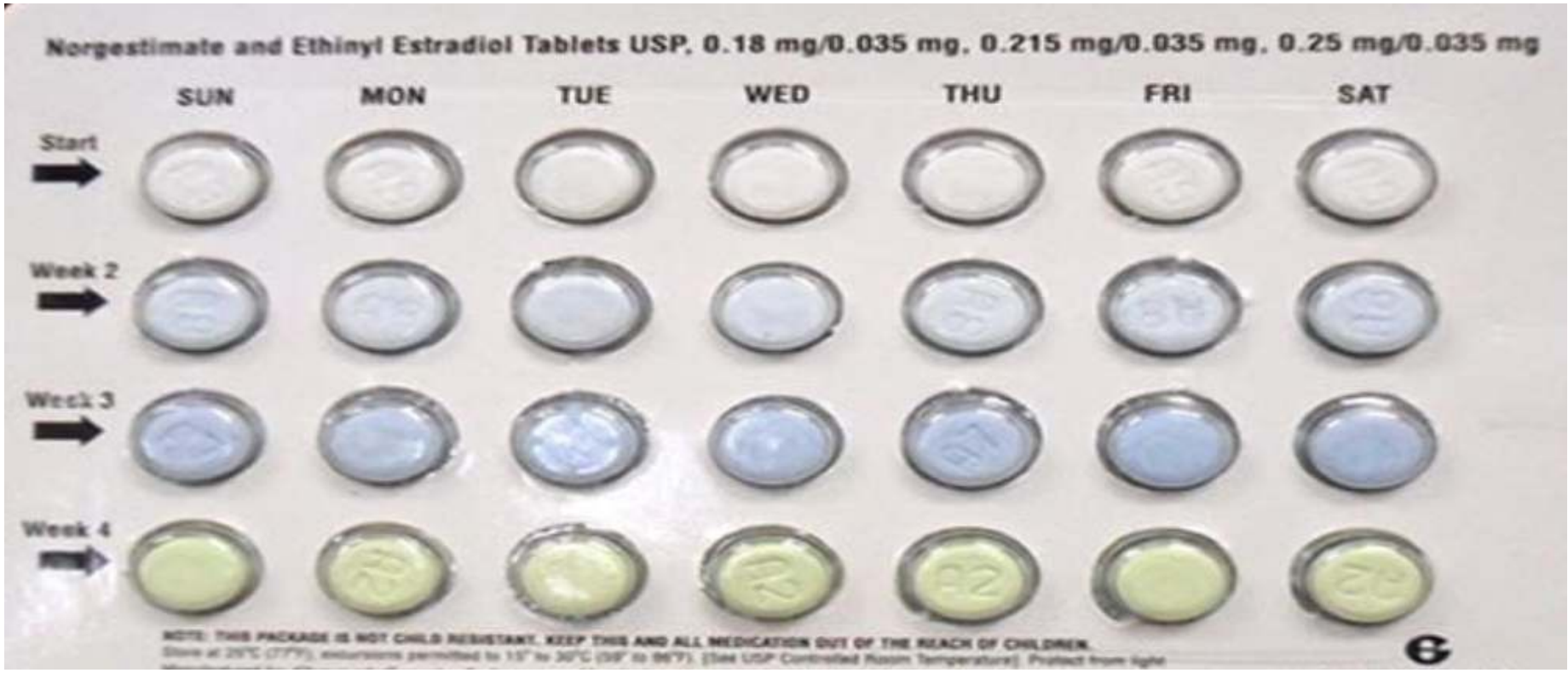
CONTRACEPTIVES

interference with ovulation is the most common pharmacologic intervention for prevention of pregnancy.

Major classes of contraceptives

1. ***Combination oral contraceptives:*** Products containing a combination of an ***estrogen and progestin*** are the most common type of oral contraceptives.

- **Monophasic** combination pills contain a constant dose of estrogen and progestin given over 21 to 24 days.
- **Triphasic** oral contraceptive products attempt to mimic the natural female cycle and most contain a constant dose of estrogen with increasing doses of progestin given over three successive 7-day periods.


Norgestimate and Ethinyl Estradiol Tablets USP, 0.18 mg/0.035 mg, 0.215 mg/0.035 mg, 0.25 mg/0.035 mg
SUN
MON
TUE
WED
THU
FRI
SAT
Start
Week 2
Week 3
Week 4
NOTE: THIS PACKAGE IS NOT CHILD RESISTANT. KEEP THIS AND ALL MEDICATION OUT OF THE REACH OF CHILDREN.

- The most common estrogen in the combination pills is **ethinyl estradiol**.
- The most common progestins are [norethindrone, desogestrel and drospirenone]
- With most oral contraceptives, active pills are taken for 21 to 24 days, followed by 4 to 7 days of placebo, for a total regimen of 28 days. Withdrawal bleeding occurs during the hormone-free (placebo) interval.

2. ***Transdermal patch***: An alternative to combination oral contraceptives is a transdermal containing a combination of an **estrogen and progestin** .

One contraceptive patch is applied each week for 3 weeks to the abdomen, upper torso, or buttock. No patch is applied during the 4th week, and withdrawal bleeding occurs.



3. ***Vaginal ring:*** containing ethinyl **estrogen and progestin.**

4. ***Progestin-only pills***: Products containing a progestin only, (called a “mini-pill”), are taken daily on a continuous schedule.
They are less effective than combination products
The progestin-only pill may be used for patients who are

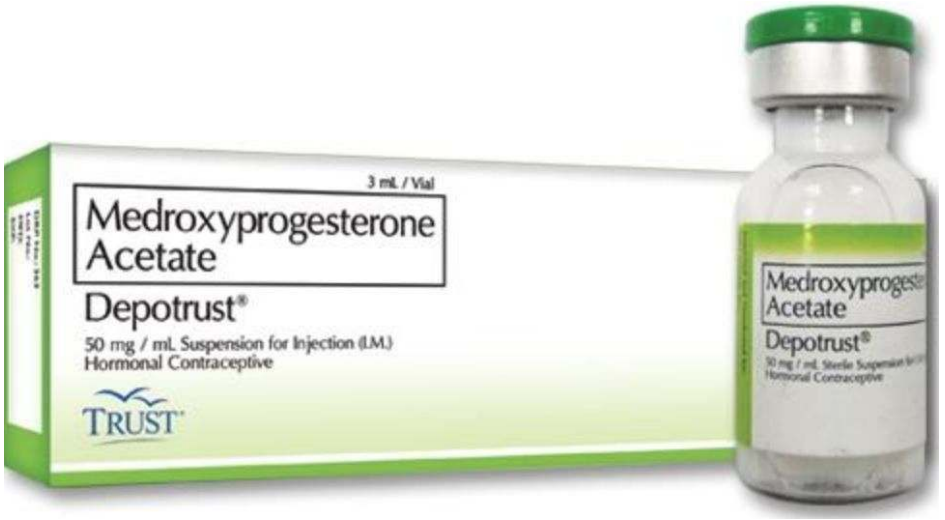
- ✓ Breast-feeding (unlike estrogen, progestins do not have an effect on milk production),
- ✓ Intolerant to estrogen,
- ✓ Smokers



5. Injectable progestin: **Medroxyprogesterone acetate** is a contraceptive that is administered via intramuscular or subcutaneous injection **every 3 months.**

adverse effect:

- weight gain is common
- **return to fertility may be delayed for several months after discontinuation**.
- **Osteoporosis and/or fractures.**



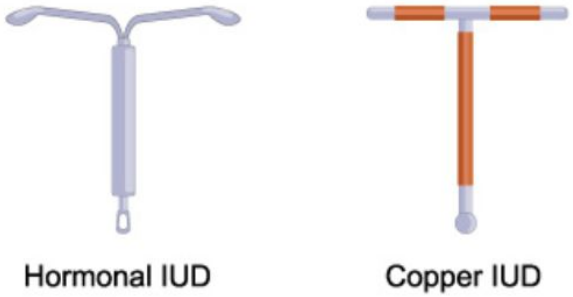
6. Progestin implants: After subdermal placement, the implant offers contraception for approximately 3 years.





7. Progestin intrauterine device: A **Progestin-releasing intrauterine system** offers a highly effective method of contraception for 3 to 5 years depending on the system. It should be avoided in patients with

1. **Pelvic inflammatory disease**
2. **History of ectopic pregnancy.**



8. Postcoital contraception: Postcoital or emergency contraception reduces the probability of pregnancy after an episode of coitus without effective contraception

- Emergency contraception uses high doses of **levonorgestrel** (preferred) or high doses of **ethinyl estradiol plus levonorgestrel.**
- For maximum effectiveness, emergency contraception should be taken as soon as possible after unprotected intercourse and preferably within 72 hours.



- An alternative emergency contraceptive is the selective progesterone modulator ***ulipristal***. It is indicated for emergency contraception within 5 days of unprotected intercourse.

Mechanism of action of contraceptives:

- Estrogen provides a negative feedback on the release of LH and FSH by the pituitary gland, thus preventing ovulation.
- Progestin thickens the cervical mucus, thus hampering the transport of sperm.
- Withdrawal of the progestin stimulates menstrual bleeding during the placebo week.

Adverse effects:

- ✓ The most common adverse effects with estrogens are *breast fullness, fluid retention, headache, and nausea. Increased blood pressure may also occur.*
- ✓ Progestins may be associated with *depression, hirsutism and acne*.
- ✓ The incidence of cervical cancer may be increased with oral contraceptives
- ✓ Rarely: more in women over 35 years
- Thromboembolism.
- Myocardial infarction, and stroke.
- Cholecystitis and gall stones.

- **Contraindications**

1. Hypertension or ischemic heart disease (IHD).
2. History of embolism, thrombosis or cerebral hemorrhage.
3. History of cancer breast or estrogen-dependent neoplasm.
4. Migraine headache.
5. Chronic liver disease and gall stones.
6. Diabetes mellitus.
7. Obese, smokers
8. Pregnancy.
9. Depression.

Causes of failure of contraceptive pills:

- If taken with enzyme inducers e.g. rifampin, phenytoin, etc.
- If woman taking broad spectrum antibiotics e.g tetracyclines.
- Paraffin oil (laxative) ↓ intestinal absorption of contraceptive pills.

- **Lippincott's Illustrated Reviews Pharmacology 6th edition**